

MFML AND MFDS

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AIMLZC416, DSECLZC416- MFML, MFDS

Webinar: #1

AGENDA

Problems on

- System of linear equations
- Vector spaces
- Inner Product Space
- Cholesky Factorization

Problem 1

You are given a system of three linear equations as shown below:

$$\alpha x + 2y + 6z = b_1$$
$$\alpha x + \alpha y + 5z = b_2$$
$$\alpha x + \alpha y + \alpha z = b_3$$

In each of the equations, you can observe the presence of an unknown constant α. Derive the three possible values of α for which the given linear system will fail to have three pivots. Provide a valid reason in each case.

CONT

Consider the coefficient matrix

$$\mathbf{A} = \begin{bmatrix} \alpha & 2 & 6 \\ \alpha & \alpha & 5 \\ \alpha & \alpha & \alpha \end{bmatrix}.$$

$R_2 = R_2 - R_1$

$R_3 = R_3 - R_1$

$$\begin{bmatrix} \alpha & 2 & 6 \\ 0 & \alpha-2 & -1 \\ 0 & \alpha-2 & \alpha-6 \end{bmatrix}.$$

To convert it into Row Echelon Form (REF), perform row operations:

$$\mathbf{A} \xrightarrow{R_2 \leftarrow R_2 - R_1,\ R_3 \leftarrow R_3 - R_1} \begin{bmatrix} \alpha & 2 & 6 \\ 0 & \alpha-2 & -1 \\ 0 & \alpha-2 & \alpha-6 \end{bmatrix}$$

$\alpha - 6 - (-1)$

$\alpha - 5$

$R_3 = R_3 - R_2$

$$\begin{bmatrix} \alpha & 2 & 6 \\ 0 & \alpha-2 & -1 \\ 0 & 0 & \alpha-5 \end{bmatrix}.$$

$\alpha = 0$

$\alpha - 2 = 0, \alpha = 2$

$\alpha - 5 = 0, \alpha = 5$

$\alpha \in \{0, 2, 5\}$

Continue row reduction:

CONT

$$\xrightarrow{R_3 \leftarrow R_3 - R_2} \begin{bmatrix} \alpha & 2 & 6 \\ 0 & \alpha - 2 & -1 \\ 0 & 0 & \alpha - 5 \end{bmatrix}.$$

Hence, the REF of **A** is

$$\begin{bmatrix} \alpha & 2 & 6 \\ 0 & \alpha - 2 & -1 \\ 0 & 0 & \alpha - 5 \end{bmatrix}.$$

Clearly, **A** will not have 3 pivots if

$$\alpha = 0, \quad \text{or } \alpha - 2 = 0, \quad \text{or } \alpha - 5 = 0.$$

$$\boxed{\alpha = 0,\ 2,\ 5}.$$

PROBLEM 2

Let $\mathbf{A} = \begin{bmatrix} -3 & 6 & -1 & 1 & -7 \\ 1 & -2 & 2 & 3 & -1 \\ 2 & -4 & 5 & 8 & -4 \end{bmatrix}$

(a) Using elementary row operations, write the matrix in its REF.

(b) Let $\mathbf{V}$ be a vector subspace spanned by the columns of matrix $\mathbf{A}$. Give a set of vectors that form the basis for the vector space $\mathbf{V}$. What is the number of vectors in this basis?

(c) Let $\mathbf{V}$ be a vector subspace spanned by vectors $\mathbf{x}$, such that $\mathbf{Ax} = 0$. Give a set of vectors that form the basis for the vector space $\mathbf{V}$. What is the number of vectors in this basis?

(d) Give the set of linearly independent rows of $\mathbf{A}$. What is the number of vectors in this set?

$$A = \begin{bmatrix} -3 & 6 & -1 & 1 & -7 \\ 1 & -2 & 2 & 3 & -1 \\ 2 & -4 & 5 & 8 & -4 \end{bmatrix}$$

$R_1 \leftrightarrow R_2$

$$\sim \begin{bmatrix} 1 & -2 & 2 & 3 & -1 \\ -3 & 6 & -1 & 1 & -7 \\ 2 & -4 & 5 & 8 & -4 \end{bmatrix}$$

$[A : 0]$

$R_2 = R_2 + 3R_1, \quad R_3 = R_3 - 2R_1$

$$\sim \begin{bmatrix} 1 & -2 & 2 & 3 & -1 \\ 0 & 0 & 5 & 10 & -10 \\ 0 & 0 & 1 & 2 & -2 \end{bmatrix}$$

$R_3 = 5R_3 - R_2$

$$\sim \begin{bmatrix} 1 & -2 & 2 & 3 & -1 \\ 0 & 0 & 5 & 10 & -10 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Row-echelon form.

C_1, C_3

$C_2 = -2 \times C_1$

$C_1 \rightarrow$ indep

$C_1, C_3 \rightarrow$

$$\text{Colspace}(A) = \text{dim} = 2, \quad \left\{ \begin{bmatrix} -3 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} -1 \\ 2 \\ 5 \end{bmatrix} \right\}$$

$Ax = 0$

$$\begin{bmatrix} 1 & -2 & 2 & 3 & -1 \\ 0 & 0 & 5 & 10 & -10 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$x_1 - 2x_2 + 2x_3 + 3x_4 - x_5 = 0$ — (1)

$5x_3 + 10x_4 - 10x_5 = 0$ — (2)

$\rho(A) = 4$

$x_1 + x_2 + x_3 = 5$

$3 - 1 = 2$

Soln of $Ax = 0$

num of free var = num of var − num of eqns

$= 5 - 2 = 3$

Free variables are x_2, x_4, x_5

$x_2 = r,\ x_4 = s,\ x_5 = t;\ r, s, t \in \mathbb{R}$

(2) $\Rightarrow$ $5x_3 = -10x_4 + 10x_5$

$x_3 = \frac{-10}{5}s + \frac{10}{5}t$

$x_3 = -2s + 2t$

$x_1 = 2r + s - 3t$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = r\begin{bmatrix} 1 \\ 0 \\ \cdot \\ \cdot \\ \cdot \end{bmatrix} + s\begin{bmatrix} 0 \\ 1 \\ \cdot \\ \cdot \\ \cdot \end{bmatrix}$$

$$
\begin{aligned}
x_1 &= 2r + s - 3t \\
x_2 &= r \\
x_3 &= -2s + 2t \\
x_4 &= s \\
x_5 &= t
\end{aligned}
\Bigg\}
$$

$$
\begin{bmatrix} 2r+s-3t \\ r \\ -2s+2t \\ s \\ t \end{bmatrix} = r\begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + s\begin{bmatrix} 1 \\ 0 \\ -2 \\ 1 \\ 0 \end{bmatrix} + t\begin{bmatrix} -3 \\ 0 \\ 2 \\ 0 \\ 1 \end{bmatrix}
$$

Basis $\Rightarrow$ $\left\{ \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ -2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ 0 \\ 2 \\ 0 \\ 1 \end{bmatrix} \right\}$

dim $= 3$.

SOLUTION

(a)

$$\mathbf{A} = \begin{bmatrix} -3 & 6 & -1 & 1 & -7 \\ 1 & -2 & 2 & 3 & -1 \\ 2 & -4 & 5 & 8 & -4 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{bmatrix} 1 & -2 & 2 & 3 & -1 \\ -3 & 6 & -1 & 1 & -7 \\ 2 & -4 & 5 & 8 & -4 \end{bmatrix}$$

$$\xrightarrow{R_2 \leftarrow R_2 + 3 * R_1, R_3 \leftarrow R_3 - 2 * R_1, R_2 \leftarrow R_2/5, R_3 \leftarrow R_3 - R_2, R_1 \leftarrow R_1 - 2 * R_2}$$

$$\begin{bmatrix} 1 & -2 & 0 & -1 & 3 \\ 0 & 0 & 1 & 2 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

NOTE: REF is not unique.

CONT

(b) column number 1 and column number 3 have pivot.
Basis of column space is:

$$\begin{bmatrix} -3 \\ 1 \\ 2 \end{bmatrix} \quad \begin{bmatrix} -1 \\ 2 \\ 5 \end{bmatrix}$$

Rank of col space = 2

CONT

(c) From REF, the null space is:

$$\begin{bmatrix} x \\ r \\ y \\ s \\ t \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} r + \begin{bmatrix} 1 \\ 0 \\ -2 \\ 1 \\ 0 \end{bmatrix} s + \begin{bmatrix} -3 \\ 0 \\ 2 \\ 0 \\ 1 \end{bmatrix} t$$

Rank of null space = 3

(d) From REF, row 1 and row 2 form basis of row space.
Rank of row space = 2

PROBLEM 3

Suppose $\mathbf{A} = [\mathbf{C}_1, \mathbf{C}_2, \mathbf{C}_3, \mathbf{C}_4]$, where $\mathbf{C}_m \in \mathbb{R}^4$, $m = 1, 2, 3, 4$ are columns of $\mathbf{A}$. It is known that rank$(\mathbf{A}) = 2$ and $\mathbf{C}_2 = 3\mathbf{C}_1$ and $\mathbf{C}_4 = 2\mathbf{C}_1 + 3\mathbf{C}_3$. If a particular solution of $\mathbf{Ax} = \mathbf{b}$ is $[1, 0, 1, 0]^T$, then

(a) Find the general solution of $\mathbf{Ax} = \mathbf{b}$.

(b) Find $\mathbf{b}$ if RREF$(\mathbf{A}) = \mathbf{A}$.

SOLUTION

(a) Now $\mathbf{A} = [\mathbf{C}_1, \mathbf{C}_2, \mathbf{C}_3, \mathbf{C}_4]$ where $\mathbf{C}_m \in \mathbb{R}^4, m = 1, 2, 3, 4$ are columns of **A**. It is known that rank$(\mathbf{A}) = 2$

$\mathbf{C}_2 = 3\mathbf{C}_1 \Rightarrow 3\mathbf{C}_1 - \mathbf{C}_2 + 0\mathbf{C}_3 + 0\mathbf{C}_4 = \mathbf{0}$

$$\Rightarrow [\mathbf{C}_1, \mathbf{C}_2, \mathbf{C}_3, \mathbf{C}_4] \begin{bmatrix} 3 \\ -1 \\ 0 \\ 0 \end{bmatrix} = \mathbf{0}$$

$\mathbf{C}_4 = 2\mathbf{C}_1 + 3\mathbf{C}_3 \Rightarrow 2\mathbf{C}_1 + 0\mathbf{C}_2 + 3\mathbf{C}_3 - 1\mathbf{C}_4 = \mathbf{0}$

$$\Rightarrow [\mathbf{C}_1, \mathbf{C}_2, \mathbf{C}_3, \mathbf{C}_4] \begin{bmatrix} 2 \\ 0 \\ 3 \\ -1 \end{bmatrix} = \mathbf{0}$$

Thus, the general solution of $\mathbf{Ax} = \mathbf{b}$ is given by

$$\left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix} + \lambda_1 \begin{bmatrix} 3 \\ -1 \\ 0 \\ 0 \end{bmatrix} + \lambda_2 \begin{bmatrix} 2 \\ 0 \\ 3 \\ -1 \end{bmatrix} \mid \forall \lambda_1, \lambda_2 \in \mathbb{R} \right\}$$

(b) $\mathbf{A} = [\mathbf{C}_1, \mathbf{C}_2, \mathbf{C}_3, \mathbf{C}_4] = [\mathbf{C}_1, 3\mathbf{C}_1, \mathbf{C}_3, 2\mathbf{C}_1 + 3\mathbf{C}_3]$ and $\text{rank}(\mathbf{A}) = 2$. Therefore $\mathbf{C}_1, \mathbf{C}_3$ are pivot columns & $\text{RREF}(\mathbf{A}) = \mathbf{A}$.

$\Rightarrow \mathbf{C}_1 = [1, 0, 0, 0]^T$ and $\mathbf{C}_3 = [0, 1, 0, 0]^T$

$\Rightarrow \mathbf{C}_2 = [3, 0, 0, 0]^T$ and $\mathbf{C}_4 = [2, 3, 0, 0]^T$

$$\Rightarrow \mathbf{A} = \begin{bmatrix} 1 & 3 & 0 & 2 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \text{Therefore } \mathbf{b} = \mathbf{A}[1, 0, 1, 0]^T = [1, 1, 0, 0]^T$$

PROBLEM 4

Consider $M = \{\mathbf{A} \in \mathbb{R}^{2\times 2} | \mathbf{A} = -\mathbf{A}^T\}$.

(a) Prove that M is a subspace of vectore space V, where V is set of all 2×2 real matrices.

(b) Prove or disprove $\left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$ is a basis for M defined in above.

(c) Prove or disprove that $\left\{ \begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 5 & -5 \\ 5 & 5 \end{bmatrix}, \begin{bmatrix} 0 & -3 \\ 3 & 1 \end{bmatrix} \right\}$ is a linearly independent set.

SOLUTION

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(a) Clearly the zero matrix of order 2, $\mathbf{0} \in M \Rightarrow M \neq \phi$.

Let $\mathbf{A}, \mathbf{B} \in M, k \in \mathbb{R}$

$\Rightarrow \mathbf{A} = -\mathbf{A}^T, \mathbf{B} = -\mathbf{B}^T$

$\Rightarrow \mathbf{A} + \mathbf{B} = -\mathbf{A}^T - \mathbf{B}^T = -(\mathbf{A} + \mathbf{B})^T$

$\Rightarrow \mathbf{A} + \mathbf{B} \in M$

$k\mathbf{A} = k(-\mathbf{A}^T) = (-k)\mathbf{A}^T = -(k\mathbf{A})^T$

So, M is a subspace.

(b) Clearly $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$ does not belong to M and hence cannot be a basis of M.

(c) Consider $a\begin{bmatrix}1 & -1\\ 1 & 0\end{bmatrix} + b\begin{bmatrix}5 & -5\\ 5 & 5\end{bmatrix} + c\begin{bmatrix}0 & -3\\ 3 & 1\end{bmatrix} = \begin{bmatrix}0 & 0\\ 0 & 0\end{bmatrix}$

$$\begin{aligned} \Rightarrow a + 5b &= 0 \\ -a - 5b - 3c &= 0 \\ a + 5b + 3c &= 0 \\ 5b + c &= 0 \end{aligned}$$

$\Rightarrow a = b = c = 0$. Hence it a linearly independent set.

PROBLEM 5

Let $\mathbf{A} = \begin{bmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{bmatrix}$.

Then define $\langle \mathbf{x}, \mathbf{y} \rangle_{\mathbf{A}} = \mathbf{x}^T \mathbf{A} \mathbf{y}$ where $\mathbf{x}, \mathbf{y} \in \mathbb{R}^4$.

(a) Prove that $\langle ., . \rangle_{\mathbf{A}}$ is an inner product on $\mathbb{R}^4$.

(b) Find all possible $\mathbf{y} = [y_1, y_2, y_3, y_4]^T$ such that $\langle \mathbf{x}, \mathbf{y} \rangle_{\mathbf{A}} = 0$ where $\mathbf{x} = [1, 0, 0, 0]^T$

SOLUTION

(a) Clearly $\mathbf{A} = \mathbf{A}^T$ and hence $\mathbf{A}$ is symmetric. Now

$$\begin{aligned}\mathbf{x}^T\mathbf{A}\mathbf{x} &= 2x_1^2 - 2x_1x_2 + 2x_2^2 - 2x_2x_3 + 2x_3^2 - 2x_3x_4 + 2x_4^2 \\ &= x_1^2 + (x_1 - x_2)^2 + (x_2 - x_3)^2 + (x_3 - x_4)^2 + x_4^2 > 0\end{aligned}$$

if $\mathbf{x} \neq 0$.

Thus $\mathbf{A}$ is symmetric positive definite. Hence by theorem $\langle \mathbf{x}, \mathbf{y} \rangle_{\mathbf{A}} = \mathbf{x}^T\mathbf{A}\mathbf{y}$ is an inner product.

(b) If $y = [y_1, y_2, y_3, y_4]^T$ and $\mathbf{x} = [1, 0, 0, 0]^T$ we want $\langle \mathbf{x}, \mathbf{y} \rangle_{\mathbf{A}} = 0$

$\Rightarrow 2y_1 - y_2 = 0 \Rightarrow 2y_1 = y_2$.

Thus $\mathbf{y} = [y_1, y_2, y_3, y_4]^T$ with $2y_1 = y_2$, $\langle \mathbf{x}, \mathbf{y} \rangle_{\mathbf{A}} = 0$.

PROBLEM 6

Consider the positive definite matrix $A \in \mathbb{R}^{3\times 3}$ defined as:

$$A = \begin{bmatrix} 9 & 6 & 3 \\ 6 & 8 & 12 \\ 3 & 12 & 75 \end{bmatrix}$$

(i) Derive a lower triangular matrix L so that A can be factorized as $A = LL^T$. Derive all the three eigenvalues of L.

(ii) Let $x \in \mathbb{R}^3$ be a nonzero vector. Define $z = L^T x$ where $L \in \mathbb{R}^{3\times 3}$ satisfies the equality $A = LL^T$. Prove that z can never be the zero vector.

SOLUTION

Given

$$A = \begin{bmatrix} 9 & 6 & 3 \\ 6 & 8 & 12 \\ 3 & 12 & 75 \end{bmatrix}$$

Since A is positive definite, it admits a Cholesky factorization $A = LL^T$ where L is lower triangular.

$$L = \begin{bmatrix} \ell_{11} & 0 & 0 \\ \ell_{21} & \ell_{22} & 0 \\ \ell_{31} & \ell_{32} & \ell_{33} \end{bmatrix}$$

Then

$$LL^T = \begin{bmatrix} \ell_{11}^2 & \ell_{11}\ell_{21} & \ell_{11}\ell_{31} \\ \ell_{21}\ell_{11} & \ell_{21}^2 + \ell_{22}^2 & \ell_{21}\ell_{31} + \ell_{22}\ell_{32} \\ \ell_{31}\ell_{11} & \ell_{31}\ell_{21} + \ell_{32}\ell_{22} & \ell_{31}^2 + \ell_{32}^2 + \ell_{33}^2 \end{bmatrix}$$

Equating with A:

$$\ell_{11}^2 = 9 \Rightarrow \ell_{11} = 3$$

$$3\ell_{21} = 6 \Rightarrow \ell_{21} = 2$$

$$3\ell_{31} = 3 \Rightarrow \ell_{31} = 1$$

$$\ell_{21}^2 + \ell_{22}^2 = 8 \Rightarrow 4 + \ell_{22}^2 = 8 \Rightarrow \ell_{22} = 2$$

$$\ell_{21}\ell_{31} + \ell_{22}\ell_{32} = 12 \Rightarrow 2 + 2\ell_{32} = 12 \Rightarrow \ell_{32} = 5$$

$$\ell_{31}^2 + \ell_{32}^2 + \ell_{33}^2 = 75 \Rightarrow 1 + 25 + \ell_{33}^2 = 75 \Rightarrow \ell_{33} = 7$$

$$L = \begin{bmatrix} 3 & 0 & 0 \\ 2 & 2 & 0 \\ 1 & 5 & 7 \end{bmatrix}$$

CONT.

Since L is triangular, its eigenvalues are its diagonal entries:

$$\lambda_1 = 3, \quad \lambda_2 = 2, \quad \lambda_3 = 7$$

(ii) Let $x \in \mathbb{R}^3$ be nonzero and define $z = L^T x$. Then

$$x^T A x = x^T L L^T x = (L^T x)^T (L^T x) = z^T z = \|z\|^2$$

Since A is positive definite,

$$x^T A x > 0 \quad \forall x \neq 0$$

Hence

$$\|z\|^2 > 0 \Rightarrow z \neq 0$$

$z = L^T x$ can never be the zero vector if $x \neq 0$.

Problem 7

(a) Assume that you are given a matrix $A \in \mathbb{R}^{n \times n}$ where $A = [a_{ij}]$ $(1 \leq i, j \leq n)$. The matrix A has the following interesting property:

$$a_{i1} + a_{i2} + \cdots + a_{in} = 0 \quad \text{for each of } i = 1, 2, \ldots, n.$$

(i) Can zero be one of the eigenvalues of this matrix? Give reasoning for your answer.

(ii) Derive the determinant of this matrix.

(iii) Derive at least one eigenvector–eigenvalue pair of this matrix.

SOLUTION

(a) Given $A = [a_{ij}] \in \mathbb{R}^{n \times n}$ such that

$$a_{i1} + a_{i2} + \cdots + a_{in} = 0 \quad \text{for each } i = 1, 2, \ldots, n.$$

(i) Let

$$\mathbf{e} = \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}.$$

Then

$$A\mathbf{e} = \begin{bmatrix} a_{11} + a_{12} + \cdots + a_{1n} \\ a_{21} + a_{22} + \cdots + a_{2n} \\ \vdots \\ a_{n1} + a_{n2} + \cdots + a_{nn} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}.$$

SOLUTION

Hence

$$A\mathbf{e} = 0\mathbf{e}.$$

0 is an eigenvalue of A.

(ii) Since 0 is an eigenvalue, A is singular. Therefore,

$$\det(A) = 0.$$

(iii) An eigenvalue–eigenvector pair is

$$\lambda = 0, \quad v = \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}.$$

PROBLEM 8

Consider the data matrix given below

$$C = \begin{bmatrix} -5 & 10 & -5 & 5 \\ 8 & -16 & -4 & -20 \\ 4 & -8 & 5 & -3 \\ -3 & 6 & 5 & 11 \end{bmatrix}.$$

(i) Reduce the matrix C to its reduced row echelon form (RREF).

(ii) Find a basis for the subspace spanned by the columns of the matrix C.

(iii) Suppose that C is the augmented matrix of a system of linear equations $A\mathbf{x} = \mathbf{b}$. Determine whether the system is consistent. If it is consistent, find its general solution.

b) If $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4 \in \mathbb{R}^4$ and $\mathbf{v}_4$ is not a linear combination of $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$, then prove or disprove that

$$\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\}$$

forms a linearly independent set.

c) If $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4 \in \mathbb{R}^3$ and $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_4\}$ is a linearly independent set, then prove or disprove that

$$\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\}$$

forms a linearly dependent set.

SOLUTION

i) $R_1 \leftarrow \frac{R_1}{-5}$:

$$\begin{bmatrix} 1 & -2 & 1 & -1 \\ 8 & -16 & -4 & -20 \\ 4 & -8 & 5 & -3 \\ -3 & 6 & 5 & 11 \end{bmatrix}$$

$R_2 \leftarrow R_2 - 8R_1, R_3 \leftarrow R_3 - 4R_1, R_4 \leftarrow R_4 + 3R_1, R_2 \leftarrow \frac{R_2}{-12}$

$$\begin{bmatrix} 1 & -2 & 1 & -1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 8 & 8 \end{bmatrix}$$

$R_1 \leftarrow R_1 - R_2, R_3 \leftarrow R_3 - R_2, R_4 \leftarrow R_4 - 8R_2$:

$$\begin{bmatrix} 1 & -2 & 0 & -2 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

ii) Since, column 1, 3 are pivot columns,

$$\text{Basis} = \left\{[-5,\ 8,\ 4,\ -3]^T,\ [-5,\ -4,\ 5,\ 5]^T\right\}$$

iii) Given $C = [A \mid b], \Rightarrow \text{rank}(C) = \text{rank}[A \mid b]$.
Clearly, from (i),

$$\text{rank}(A) = \text{rank}[A \mid b] = 2.$$

i.e., System is consistent.
Since column 2 is a non-pivot column, x_2 is a free variable. Let $x_2 = t$.
From row 2 of rref$[A \mid b]$, we get $x_3 = 1$. From row 1, $x_1 - 2x_2 = -2 \Rightarrow x_1 = -2 + 2t$.

The general solution is

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix} + t \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}, \qquad \forall\, t \in \mathbb{R}.$$

b) False, as $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ can be a linearly dependent set, making $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\}$ a linearly dependent set.

c) Yes, as $\mathbb{R}^3$ has dimension 3, so any set containing more than 3 vectors is linearly dependent.

PROBLEM 9

a) Define

$$\langle \mathbf{x}, \mathbf{y} \rangle_A = \mathbf{x}^T A \mathbf{y} \quad \text{where } \mathbf{x}, \mathbf{y} \in \mathbb{R}^2 \quad \text{and} \quad A = \begin{bmatrix} 4 & 2 \\ 2 & 6 \end{bmatrix}.$$

(i) Prove $\langle \cdot, \cdot \rangle_A$ is an inner product on $\mathbb{R}^2$.

(ii) Find $\|\mathbf{x}\|_A$ where $\| \cdot \|_A$ is the norm induced by the inner product defined above for $\mathbf{x} = [1, 2]^T$.

b) Let A be a 2×2 symmetric real matrix with trace of A equal to 0. Can A be a symmetric positive definite matrix? Justify your answer.

c) A professor gave his student the following matrix

$$A = \begin{bmatrix} 4 & 1 & 1 \\ 1 & 3 & 1 \\ 1 & 1 & 2 \end{bmatrix}.$$

Help the student to find a lower triangular matrix L such that

$$A = LL^T.$$

a) i) Clearly $A = \begin{bmatrix} 4 & 2 \\ 2 & 6 \end{bmatrix} = A^T$

And $\det A_{11} = \det[4] = 4 > 0, \det A = 24 - 4 = 20 > 0$

$\therefore$ A is symmetric and positive definite

$\Rightarrow \langle x, y \rangle_A = x^T A y$ is aninner product on $\mathbb{R}^2$.

ii) Note that $\langle x, x \rangle_A = [1, 2] \begin{bmatrix} 4 & 2 \\ 2 & 6 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

$= [1,2] \begin{bmatrix} 8 \\ 14 \end{bmatrix} = 8+28=36$

$\therefore \quad \|x\|_A = \sqrt{\langle x, x \rangle_A} = \sqrt{36} = 6$

b) Trace of A = sum of eigenvalues = 0.

Clearly, both eigenvalues are real due to the spectral theorem. If one is positive, the other eigenvalue has to be negative, as their sum is zero.

$\therefore$ Eigenvalues are not strictly positive.

$\Rightarrow$ It is not positive definite.

c) Let L= $\begin{bmatrix} \ell_{11} & 0 & 0 \\ \ell_{21} & \ell_{22} & 0 \\ \ell_{31} & \ell_{32} & \ell_{33} \end{bmatrix}$

$$\Rightarrow LL^T = \begin{bmatrix} \ell_{11}^2 & \ell_{11}\ell_{21} & \ell_{11}\ell_{31} \\ \ell_{11}\ell_{21} & \ell_{21}^2 + \ell_{22}^2 & \ell_{21}\ell_{31} + \ell_{22}\ell_{32} \\ \ell_{11}\ell_{31} & \ell_{21}\ell_{31} + \ell_{22}\ell_{32} & \ell_{31}^2 + \ell_{32}^2 + \ell_{33}^2 \end{bmatrix}$$

$$4=\ell_{11}^2 \quad \Rightarrow \quad \ell_{11}=2, \quad 1=\ell_{11}\ell_{21} \quad \Rightarrow \quad \ell_{21}=0.5.$$

$$1=\ell_{11}\ell_{31} \quad \Rightarrow \quad \ell_{31}=0.5$$

$$3=\ell_{21}^2+\ell_{22}^2 \quad \Rightarrow \quad \ell_{22}=\sqrt{3-0.25}=\sqrt{2.75}=1.6583$$

$$1=0.5(0.5)+\sqrt{2.75}\,\ell_{32} \Rightarrow \ell_{32}=\frac{0.75}{\sqrt{2.75}}=0.4523$$

$$2=0.25+\frac{(0.75)^2}{2.75}+\ell_{33}^2, \Rightarrow \ell_{33}=\sqrt{1.75-\frac{(0.75)^2}{2.75}}=1.2432$$



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THANK YOU